WEAK CHARACTERIZATION: FINITE U AND R FIBRES OVER A POINT VERSUS A MONOMORPHISM FROM A FIELD WITH FINITE FIBRE

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Editorial note. The source marks the two fibre identifications 'clear'; both are the displayed set equalities via points-cartesian, carried as explicit obligations.

Allowed standard prerequisites (editorial).

- R-finiteness characterization. lemma-R-finite-above-x (carded L26698). Condition (1) supplies the monomorphism from a field in the class of x.
- Fibres of the presentation. Properties of Spaces, lemma points-cartesian (cross-chapter, named). Identifies the fibres with the displayed finite schemes.

Source reference. Original lemma 03JU: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 534–534. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-decent-weak-ur-finite-above-x/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $x \in |X|$. Let U be a scheme and let $\varphi : U \to X$ be an étale morphism. The following are equivalent:*

(1) *x is in the image of $|U| \to |X|$, and setting $R = U \times_X U$ the fibres of both*
$$|U| \longrightarrow |X| \quad \text{and} \quad |R| \longrightarrow |X|$$
over x are finite,

(2) *there exists a monomorphism $\mathrm{Spec}(k) \to X$ with k a field in the equivalence class of x, and the fibre product $\mathrm{Spec}(k) \times_X U$ is a finite nonempty scheme over k.*

Proof. Assume (1). This clearly implies the first condition of Lemma 03JU and hence we obtain a monomorphism $\mathrm{Spec}(k) \to X$ in the class of x. Taking the fibre product we see that $\mathrm{Spec}(k) \times_X U \to \mathrm{Spec}(k)$ is a scheme étale over $\mathrm{Spec}(k)$ with finitely many points, hence a finite nonempty scheme over k, i.e., (2) holds.

Assume (2). By assumption x is in the image of $|U| \to |X|$. The finiteness of the fibre of $|U| \to |X|$ over x is clear since this fibre is equal to $|\mathrm{Spec}(k) \times_X U|$ by

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Properties of Spaces, Lemma 03H4. The finiteness of the fibre of $|R| \to |X|$ above x is also clear since it is equal to the set underlying the scheme

$$(\mathrm{Spec}(k) \times_X U) \times_{\mathrm{Spec}(k)} (\mathrm{Spec}(k) \times_X U)$$

which is finite over k. Thus (1) holds. $\square$